

Predictive Pricing Model for Real Estate Packs

1) Context & Use Case

Leboncoin sells advertising packs (10 to 200 ads/month) to professional customers such as real estate agencies.

Agencies can either:

- Buy **packs** (monthly allowance of ads).
- Use **FIM/manual uploads** (pay-per-ad basis).
- Buy **premium options** (boost, top results, photo edits, etc.).

The pricing strategy must:

- Increase revenue through controlled price increases.
 - Limit churn (natural ~5%).
 - Reduce downsell risk.
 - Encourage migration from manual (FIM) to packs (stickier customers, more ads published).
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2) Business Goals

- **Revenue growth:** +5% portfolio revenue (capped at +18% per product).
 - **Churn guardrail:** $\leq 5\%$ (natural baseline).
 - **Pack mix management:** reduce downsell, enable upsell if revenue-positive.
 - **Migration:** move agencies from FIM/manual into packs.
 - **Volume:** maintain or grow total ads published.
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3) Data & Features

We leveraged **4 years of renewal-level data**, ~10k agencies, with the following features:

- **Prices:** pack prices, FIM price, premium price (net of discount).
- **Current state:** {Manual, Pack10, Pack20, Pack50, Pack100, Pack200}.
- **Usage:** ads published, pack fill rate, # of mandates.
- **Spend:** premium & boutique spend, total spend history.
- **Profile:** tenure, discount history, agency size.
- **Market:** competitor presence (Seloger, Bien'ici, MeilleursAgents), region.
- **Budget:**
 - $\text{affordability} = \log_{10}(\text{total_spend})$,
 - $\text{budget_pressure} = \text{total_spend} / \text{peak_spend}_{12m}$.

Data Snapshot (Two-row Records to avoid cut-off)

agency	period	current_state	choice	price_fim	price_10
A	t1	Pack20	Pack50	7.0	50
→		price_20=100	price_50=200	ExpAds=28	premium_spend=150, region=South
B	t1	Manual	Pack10	7.0	50
→		price_20=100	price_50=200	ExpAds=8	premium_spend=20, region=North

(Each record is split across two rows to fit narrow layouts.)

Reshaped (Long Format — two-row records)

agency	period	alt	chosen	price_alt	slots_alt
A	t1	Pack20	0	100	20
→				inertia=1	short=8; fim_cost_proxy=-; ExpAds=28; region=South
A	t1	Pack50	1	200	50
→				inertia=0	slack=22; fim_cost_proxy=-; ExpAds=28; region=South
A	t1	Manual	0	7.0	-
→				inertia=0	-; fim_cost_proxy=196; ExpAds=28; region=South
B	t1	Manual	0	7.0	-
→				inertia=1	-; fim_cost_proxy=56; ExpAds=8; region=North
B	t1	Pack10	1	50	10
→				inertia=0	slack=2; fim_cost_proxy=-; ExpAds=8; region=North

3. Modeling Approach

We used a **multinomial logit model** (optionally nested) to predict renewal choices.

Alternatives: {Manual, Pack10, Pack20, Pack50, Pack100, Pack200, Churn}

Utility Functions:

- **Manual :**

$$U_{\text{manual},i} = \alpha_{\text{manual}} - \beta_{\text{fim}}(\text{price}_{\text{fim}} \times \text{ExpAds}_i) + \beta_{\text{inertia}} \cdot 1\{\text{current}=\text{Manual}\} + \Gamma^T X_i + \text{Region}_i$$

- **Pack j:**

$$U_{j,i} = \alpha_j - \beta_{\text{price}} \times \text{price}_j - \beta_{\text{short}} \times \max(0, \text{ExpAds}_i - \text{Slots}_j) - \beta_{\text{slack}} \times \max(0, \text{Slots}_j - \text{ExpAds}_i) + \beta_{\text{inertia}} \cdot 1\{\text{current}=j\} + \Gamma^T X_i + \text{Region}_i$$

- **Churn :**

$$U_{\text{churn},i} = \alpha_{\text{churn}} + \beta_{\text{stress}} \cdot 1\{\% \Delta \text{ price}_{\text{current}} \text{ high}\} + \Gamma^T X_i + \text{Region}_i$$

4. Training

- **Unit:** agency × renewal
- **Dependent variable:** observed choice
- **Format:** long (1 row per alternative)
- **Estimation:** maximize log-likelihood
- **Regularization:** ensures economic interpretability ($\beta_{\text{price}} > 0$ means higher price reduces utility, $\beta_{\text{inertia}} > 0$ means agencies stick to status quo)

5. Parameter Interpretations

- α_j : baseline attractiveness of each option
- β_{price} : sensitivity to pack price
- β_{fim} : penalty from manual FIM cost
- β_{short} : penalty if expected ads exceed slots (shortage → upsell)
- β_{slack} : penalty if slots exceed expected ads (slack → downsell)
- β_{inertia} : status quo bias
- Γ : customer profile effects (usage, spend, tenure, competitor presence, affordability)
- β_{stress} : churn trigger when price hike is high

6. Results & Evaluation

We simulated new prices ($\leq +18\%$) and recomputed utilities → choice probabilities → expected revenues.

New Prices Example:

- Packs +12% → P10=€22.40, P20=€39.20, P50=€89.60

- FIM +18% → €6 → €7.08
- Premium +8% → €20 → €21.60

Agency A1 – Capacity-Constrained

- Current: Pack20, ExpAds=28, fill=92%, premium high, competitor=1
- Probabilities: P50=0.55, P20=0.35, P10=0.03, Manual=0.02, Churn=0.05
- Revenues: Pack20≈€268.6, Pack50≈€265.1
- **Expected Rev ≈ €266–267**
- 👉 Upgrade to Pack50 likely (shortage penalty).

Agency A2 – Manual Mid-Volume

- Current: Manual, ExpAds=12, low premium, no competitor
- Probabilities: Pack10=0.60, Manual=0.25, Pack20=0.10, Churn=0.05
- Revenues: Manual≈€106.6, Pack10≈€58.2, Pack20≈€60.8
- **Expected Rev ≈ €70–75**
- 👉 Migration to Pack10 likely (stability, retention).

Agency A3 – Under-Utilized, Price-Sensitive

- Current: Pack50, ExpAds=18, discount history=1, competitor=1
- Probabilities: Pack20=0.40, Pack50=0.30, Pack10=0.15, Manual=0.10, Churn=0.05
- Revenues: Pack50≈€130–135, Pack20≈€80–85, Manual≈€155–160
- **Expected Rev ≈ €110–120**
- 👉 Likely downgrade to Pack20; churn risk present.

Aggregate (toy example, 3 agencies):

- Mix shifts: A1→Pack50, A2→Pack10, A3→Pack20
- Churn ≈ 5%
- Ads published: **net positive overall**

7. Deployment Plan

- Simulation engine: recompute utilities/probabilities for new prices
- Scenario testing: revenue, churn, upsell/downsells, ads volume
- Guardrails: +18% price cap, churn ≤5%
- Regional calibration: local sensitivities
- Monitoring: compare predicted vs. actual renewals

8. Business Impact

- Projected **+5–7% revenue uplift** within constraints
- Churn maintained at ~5%

- Increased migration from manual → packs (boosting ad inventory)
 - Executives gained **confidence** in pricing rollout
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9. Key Learnings

- Behavior explained by inertia, fit (short/slack), and price sensitivity
- Manual→Pack migration driven by FIM vs Pack cost trade-off
- Non-adjacent jumps (e.g., Pack10→Pack50) triggered by shortage penalty
- Churn risk concentrated in **discount-history + competitor presence** segments